

Impulse response

$$\text{ex 1, } y'(t) + 4y(t) = 2x(t)$$

$$\Rightarrow h'(t) + 4h(t) = 2\delta(t)$$

$$\Rightarrow \lambda + 4 = 0 \Rightarrow \lambda = -4$$

$$\text{therefore, } h(t) = A \cdot e^{-4t} u(t)$$

$$h'(t) = -4A e^{-4t} u(t) + A e^{-4t} \delta(t)$$

$$\text{then, } h'(t) + 4h(t) = A e^{-4t} \delta(t) = 2\delta(t)$$

$$\Rightarrow A = 2$$

$$\text{so, } h(t) = 2e^{-4t} u(t)$$

$$\text{ex 2, } y'(t) + 4y(t) = 3x'(t) + 2x(t)$$

$$\Rightarrow h'(t) + 4h(t) = 3\delta'(t) + 2\delta(t)$$

$$\Rightarrow \lambda + 4 = 0 \Rightarrow \lambda = -4$$

$$\text{therefore, } h(t) = A \cdot e^{-4t} u(t) + B\delta(t)$$

$$h'(t) = -4A e^{-4t} u(t) + A e^{-4t} \delta(t) + 4B\delta(t) + B\delta'(t)$$

$$\text{then, } h'(t) + 4h(t) = A \cdot e^{-4t} \delta(t) = 3\delta'(t) + 2\delta(t)$$

$$\Rightarrow \begin{cases} A = -10 \\ B = 3 \end{cases}$$

$$\therefore h(t) = -10 \cdot e^{-4t} u(t) + 3\delta(t) \quad (\text{跳变})$$